MATH-1014 T10B

Tutorial 12: Power Series, Taylor & Maclaurin Series

WANG Yixiao

Department of Mathematics

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Today's Roadmap

1. Concept Review: Power Series & Taylor Expansions
2. Practice Problems
3. Additional Practice: Series Expansions & Applications

Review: Anatomy of a Power Series

A power series acts like an infinite polynomial. It is a function whose domain is the set of all x for which the series converges.

Definition & Convergence

- **General Form:** $\sum_{n=0}^{\infty} c_n(x-a)^n$ is a power series centered at a.
- **Radius of Convergence (R):** There are three possibilities: converges only at $x = a$ ($R = 0$), converges for all x ($R = \infty$), or converges if $|x - a| < R$ and diverges if $|x - a| > R$.
- **Interval of Convergence:** The specific interval of x values where it converges. You *must* check the endpoints $x = a \pm R$ separately.

Review: Calculus on Power Series

We can manipulate power series using differentiation and integration to represent new functions.

Term-by-Term Operations

If $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ has radius $R > 0$:

- **Differentiation:** $f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$.
- **Integration:** $\int f(x)dx = C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$.

Warning on Boundaries

The radius of convergence R remains exactly the same when a power series is differentiated or integrated. However, the **interval of convergence** at the endpoints may change and must be re-tested.

Review: Taylor & Maclaurin Series

How do we find a power series for a general function $f(x)$? We construct it using its derivatives.

- **Taylor Series (centered at a):** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$.
- **Maclaurin Series (centered at 0):** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.
- **Uniqueness:** The coefficients $c_n = \frac{f^{(n)}(a)}{n!}$ are uniquely determined.
- **Important Maclaurin Series:**
    - $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ $(R=1)$.
    - $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ $(R=\infty)$.
    - $\sin x = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}$ $(R=\infty)$.

Problem 1: Finding the Interval of Convergence

Problem 1.
Find the radius of convergence and interval of convergence of the series:

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$$

Strategy: Use the Ratio Test to find the radius R, then test the exact boundary values manually.

Solution 1: Ratio Test and Boundaries

Step 1: Ratio Test. Let $a_n = (-3)^n x^n / \sqrt{n+1}$.

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(-3)^{n+1} x^{n+1}}{\sqrt{n+2}} \cdot \frac{\sqrt{n+1}}{(-3)^n x^n} \right| = |3x| \lim_{n\to\infty} \sqrt{\frac{n+1}{n+2}} = |3x|$$

Converges if $|3x| < 1$, so $|x| < 1/3$. The radius of convergence is $R = 1/3$.

Step 2: Check Endpoints.

- When $x = 1/3$: $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$. Converges by the Alternating Series Test.
- When $x = -1/3$: $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}}$. This is a p-series with $p = 1/2 < 1$, so it diverges.

Conclusion: The interval of convergence is $[-1/3, 1/3)$.

Problem 2: Power Series via Geometric Manipulation

Problem 2.
Find a power series representation of the function:

$$f(x) = \frac{x^3}{x+2}$$

Hint: Do not compute derivatives for the Maclaurin formula! Force the denominator to look like the geometric series template $\frac{1}{1-r}$.

Solution 2: Geometric Series Template

Step 1: Isolate the rational part and format it.

$$\frac{1}{2+x} = \frac{1}{2} \cdot \frac{1}{1+\frac{x}{2}} = \frac{1}{2} \cdot \frac{1}{1-\left(-\frac{x}{2}\right)}$$

Step 2: Expand using $\sum r^n$.

$$\frac{1}{2+x} = \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n$$

Step 3: Multiply the x^3 back in.

$$\frac{x^3}{x+2} = x^3 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^{n+3}$$

To write in standard $\sum c_k x^k$ form, shift the index: $\sum_{n=3}^{\infty} \frac{(-1)^{n-3}}{2^{n-2}} x^n$.

Problem 3: Evaluating Sums via Calculus

Problem 3.
Find the exact sum of the following series:

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

Observation

The presence of n in the numerator alongside a geometric $1/2^n$ suggests we need to differentiate the standard geometric series, which drops an n down as a coefficient.

Solution 3: Differentiation of Power Series

Step 1: Start with the geometric series and differentiate.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1)$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$$

Step 2: Multiply by x to match the target form.

$$\frac{x}{(1-x)^2} = x\sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=1}^{\infty} n x^n$$

Step 3: Substitute $x = 1/2$.

$$\sum_{n=1}^{\infty} n\left(\frac{1}{2}\right)^n = \frac{1/2}{(1-1/2)^2} = \frac{1/2}{1/4} = \mathbf{2}$$

Problem 4: Taylor Series at $a \neq 0$

Problem 4.
Represent $f(x) = \sin x$ as the sum of its Taylor series centered at $a = \pi/3$.

Hint: When $a \neq 0$, you must calculate the derivatives manually evaluated at $x = a$ to build the series.

Solution 4: Constructing the Taylor Series

Step 1: Evaluate derivatives at $\pi/3$.

$$\begin{aligned} f(x) &= \sin x & f(\pi/3) &= \sqrt{3}/2 \\ f'(x) &= \cos x & f'(\pi/3) &= 1/2 \\ f''(x) &= -\sin x & f''(\pi/3) &= -\sqrt{3}/2 \\ f'''(x) &= -\cos x & f'''(\pi/3) &= -1/2 \end{aligned}$$

Step 2: Plug into the Taylor Series formula.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(\pi/3)}{n!}(x-\pi/3)^n$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2}\frac{(x-\pi/3)}{1!} - \frac{\sqrt{3}}{2}\frac{(x-\pi/3)^2}{2!} - \frac{1}{2}\frac{(x-\pi/3)^3}{3!} + \dots$$

Problem 5: Integral Approximation

Problem 5.
Use a power series to approximate the definite integral $\int_0^{0.1} x \arctan(3x) dx$ to six decimal places.

Strategy: We cannot easily integrate $x \arctan(3x)$ by standard means. Instead, find the Maclaurin series for $\arctan(3x)$, multiply by x, integrate term-by-term, and use the Alternating Series Estimation Theorem ($|R_n| \leq b_{n+1}$).

Solution 5: Term-by-Term Integration

Step 1: Series for $\arctan(3x)$**.** We know $\arctan u = \sum_{n=0}^{\infty}(-1)^n \frac{u^{2n+1}}{2n+1}$.

$$x\arctan(3x) = x\sum_{n=0}^{\infty}(-1)^n \frac{(3x)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty}(-1)^n \frac{3^{2n+1}x^{2n+2}}{2n+1}$$

Step 2: Integrate over $[0, 0.1]$**.**

$$\int_0^{0.1} \dots dx = \sum_{n=0}^{\infty}(-1)^n \frac{3^{2n+1}x^{2n+3}}{(2n+1)(2n+3)}\Bigg|_0^{0.1} = \sum_{n=0}^{\infty}(-1)^n \frac{3^{2n+1}(0.1)^{2n+3}}{(2n+1)(2n+3)}$$

Step 3: Estimate. Let's check the terms b_n. $b_0 = \frac{3(0.1)^3}{3} = 0.001$. $b_1 = \frac{3^3(0.1)^5}{15} = 0.000018$. $b_2 = \frac{3^5(0.1)^7}{35} \approx 0.00000069$. $b_3 = \frac{3^7(0.1)^9}{63} \approx 3.47 \times 10^{-8}$. Since $b_3 < 10^{-6}$, adding up to $n = 2$ gives 0.0009827, accurate to 6 decimal places.

Problem 6: Taylor Expansion & LCT (Advanced)

Problem 6.
Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \sin\left(\frac{1}{n}\right) \right)$$

Why this is tricky

As $n \to \infty$, $a_n \to 0 - 0 = 0$. The test for divergence fails. But how fast does it go to zero? We need **Maclaurin series** to reveal the hidden order of magnitude.

Solution 6: Revealing the Order of Growth

Step 1: Use the Maclaurin series for $\sin(x)$. We know $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$
Let $x = \frac{1}{n}$. For large n (small x), we have:

$$\sin\left(\frac{1}{n}\right) \approx \frac{1}{n} - \frac{1}{6n^3}$$

Step 2: Substitute back into a_n.

$$a_n = \frac{1}{n} - \left(\frac{1}{n} - \frac{1}{6n^3} + \text{higher order terms}\right) \approx \frac{1}{6n^3}$$

Step 3: Limit Comparison Test. Let $b_n = \frac{1}{n^3}$.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{n} - \sin(1/n)}{1/n^3} \overset{\text{L'H}}{=} \cdots = \frac{1}{6} > 0$$

Since $\sum \frac{1}{n^3}$ converges ($p = 3 > 1$), the series **converges**.

Problem 7: Advanced Interval of Convergence

Problem 7. (From Part G, Q36)
Find the interval of convergence for the power series:

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(x-2)^n$$

Strategy: Apply the Ratio Test to find the radius of convergence, then carefully test both boundary points using the Alternating Series Test or p-series test.

Solution 7: Testing the Boundaries

Step 1: Ratio Test. Let $a_n = \frac{(-1)^n (x-2)^n}{(n+2)4^n}$.

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(x-2)^{n+1}}{(n+3)4^{n+1}} \cdot \frac{(n+2)4^n}{(x-2)^n} \right| = \frac{|x-2|}{4} \lim_{n\to\infty} \frac{n+2}{n+3} = \frac{|x-2|}{4}$$

For absolute convergence, $\frac{|x-2|}{4} < 1 \implies |x-2| < 4 \implies -2 < x < 6$.

Step 2: Check Endpoints.

- $x = 6$: $\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(4)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+2}$. Converges (AST).
- $x = -2$: $\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(-4)^n = \sum_{n=0}^{\infty} \frac{1}{n+2}$. Diverges (Harmonic series limit comparison).

Conclusion: The interval of convergence is $(-2, 6]$.

Problem 8: Maclaurin Series Coefficients

Problem 8. (From Part G, Q37)
Find the coefficient of the x^5 term in the Maclaurin series expansion of:

$$f(x) = \ln(2 + x^2)^2$$

Hint: Use logarithm properties to simplify the function first, then relate it to the standard expansion $\ln(1 + u)$. Think about the parity (even/odd) of the resulting powers.

Solution 8: Parity and Expansions

Step 1: Simplify using Log Properties.

$$f(x) = 2\ln(2+x^2) = 2\ln\left(2\left(1+\frac{x^2}{2}\right)\right) = 2\ln(2) + 2\ln\left(1+\frac{x^2}{2}\right)$$

Step 2: Apply standard Maclaurin Series. We know that $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$
Let $u = \frac{x^2}{2}$.

$$f(x) = 2\ln(2) + 2\left(\left(\frac{x^2}{2}\right) - \frac{1}{2}\left(\frac{x^2}{2}\right)^2 + \frac{1}{3}\left(\frac{x^2}{2}\right)^3 - \dots\right)$$

Step 3: Identify the x^5 coefficient. Notice that substituting $u = x^2/2$ guarantees that every term in the expansion will have an **even power** of x (e.g., x^2, x^4, x^6).
Therefore, there is no x^5 term. The coefficient of x^5 is 0.

Problem 9: The Binomial Series

Problem 9. (From Part G, Q38)
Find the coefficient of the x^4 term in the Maclaurin series expansion of the composite function:

$$f(x) = (1 + 8x^2)^{3/2}$$

Strategy: Use the generalized Binomial Series expansion $(1+u)^k = 1 + ku + \frac{k(k-1)}{2!}u^2 + \dots$

Solution 9: Applying the Binomial Theorem

Step 1: Set up the Binomial Series. Let $u = 8x^2$ and $k = 3/2$. The expansion is:

$$(1+u)^k = 1 + ku + \frac{k(k-1)}{2!}u^2 + \frac{k(k-1)(k-2)}{3!}u^3 + \dots$$

Step 2: Locate the required power. We want the coefficient of x^4. Since $u = 8x^2$, the x^4 term will come from the u^2 term in the expansion.

Step 3: Calculate the coefficient.

$$\text{Term with } u^2 = \frac{(3/2)(3/2-1)}{2}(8x^2)^2$$

$$= \frac{(3/2)(1/2)}{2}(64x^4) = \frac{3/4}{2}(64x^4) = \frac{3}{8}(64x^4) = \mathbf{24}x^4$$

The coefficient of the x^4 term is 24.

Problem 10: Differentiation of a Power Series

Problem 10. (From Part G, Q36c)
Given the power series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(x-2)^n$, define a new function:

$$H(x) = (x-2)^2 f(x)$$

Calculate the exact value of the derivative $H'(4)$.

Hint: Multiply the $(x-2)^2$ into the series to create a cleaner sum. Differentiate term-by-term, and recognize the resulting geometric series.

Solution 10: Recognizing Geometric Forms

Step 1: Absorb the term and differentiate.

$$H(x) = (x-2)^2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(x-2)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(x-2)^{n+2}$$

$$H'(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)4^n}(n+2)(x-2)^{n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}(x-2)^{n+1}$$

Step 2: Factor out common terms to reveal the geometric series.

$$H'(x) = (x-2)\sum_{n=0}^{\infty} \left(\frac{-(x-2)}{4}\right)^n$$

This is a geometric series with common ratio $r = -(x-2)/4$.

Step 3: Evaluate at $x = 4$. When $x = 4$, the ratio is $r = -(4-2)/4 = -1/2$. Since $|r| < 1$, we can sum it:

$$H'(4) = (4-2)\frac{1}{1-(-1/2)} = 2\left(\frac{1}{3/2}\right) = 2\left(\frac{2}{3}\right) = \mathbf{\frac{4}{3}}$$

Problem 11: Exact Sums via Splitting

Problem 11. (From Part E, Q26a)
Evaluate the exact sum of the following series:

$$\sum_{n=1}^{\infty} \frac{4(3^{n+1}) + 2^{n+1}}{6^n}$$

Strategy: Break the single fraction into two separate exponential fractions. Simplify the bases to reveal two standard geometric series, then apply the sum formula $S = \frac{a}{1-r}$.

Solution 11: Sum of Geometric Series

Step 1: Split the series and simplify exponents.

$$\sum_{n=1}^{\infty} \frac{4(3^n \cdot 3) + (2^n \cdot 2)}{6^n} = \sum_{n=1}^{\infty} \left(12\frac{3^n}{6^n} + 2\frac{2^n}{6^n} \right)$$

$$= 12\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n + 2\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$$

Step 2: Evaluate each geometric series. For $\sum_{n=1}^{\infty}(1/2)^n$, the first term is $a = 1/2$, ratio $r = 1/2$. Sum $= \frac{1/2}{1-1/2} = 1$.

For $\sum_{n=1}^{\infty}(1/3)^n$, the first term is $a = 1/3$, ratio $r = 1/3$. Sum $= \frac{1/3}{1-1/3} = 1/2$.

Step 3: Combine the results.

$$\text{Total Sum } = 12(1) + 2\left(\frac{1}{2}\right) = 12 + 1 = \mathbf{13}$$

THANK YOU!